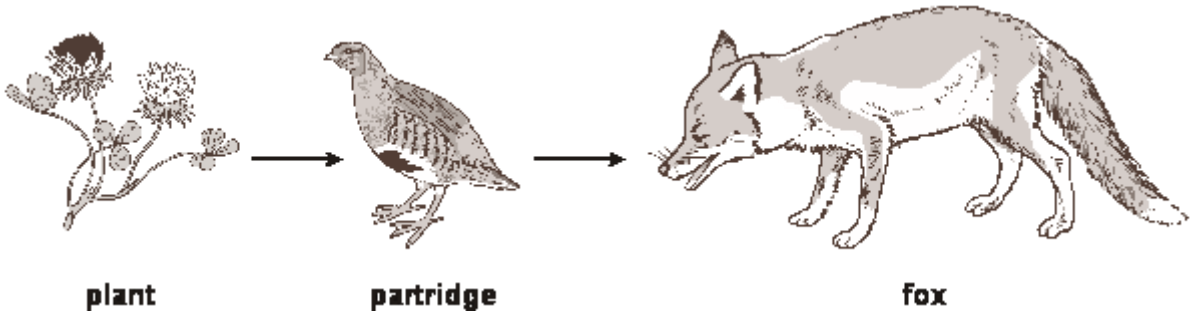
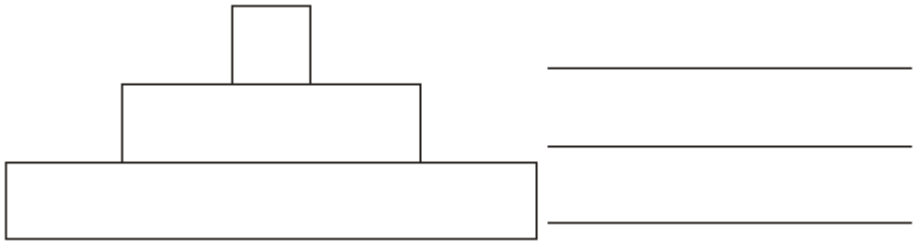
Q1. The drawings show part of a farmland food chain.



not to scale

(a) A pyramid of numbers represents the number of organisms at each stage in a food chain.

On each line by the pyramid of numbers below, write the name of the correct organism from the food chain above.



1 mark

(b) Partridges feed mainly on insects and wild plants (weeds).

Some farmers spray their crops with chemicals to kill insects and weeds.

How would this affect the number of foxes?

.....

Explain your answer.

.....

.....

1 mark

(c) Partridges build their nests on the ground among plants.
They lay up to 18 eggs in the nest.

Suggest why partridges need to lay so many eggs.

.....

.....

1 mark

(d) Some farmers leave a strip of land around the edge of each field which they do **not** spray with chemicals.

Suggest **two** reasons why this will lead to an increase in the number of partridges on these farms.

1 .....

.....

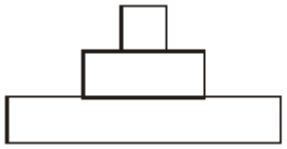
2 .....

.....

2 marks
maximum 5 marks

Q1.

(a) •



*answers must be in the correct order **all three** answers are required for the mark*

1 (L5)

(b) • fewer foxes **or** they decrease

accept 'there would be less'

accept 'they would drop or go down'

***both** the answer and the correct explanation are required for the mark*
credit both correct answers written in the same space
'foxes die' is insufficient

any **one** from

- fewer partridges (for foxes to eat)
- the poison gets into the food chain
- they move elsewhere to find food

accept 'less food'
accept 'partridges would be poisoned'
accept 'some die of starvation'

*do **not** accept 'the chemicals poison the foxes'*
*do **not** accept 'they all die of starvation'*
*do **not** accept 'they have nothing to eat'*

1 (L6)

(c) any **one** from

- greater chance of survival
- more would survive

accept 'the more eggs they lay the more partridges there will be'

*accept 'some eggs will be damaged **or** eaten'*

accept 'some eggs will not hatch'

*accept 'young partridges **or** chicks may be eaten **or** trampled on **or** die'*

*accept 'so they will not become extinct **or** die out'*

'partridges will die' is insufficient

1 (L6)

(d) any **two** from

- there would be a greater variety of plants

- more plants **or** weeds
- more insects
- more nesting places
- more cover
- the partridges are not poisoned

accept 'there would be plants to eat'

accept 'there would be insects to eat'

accept 'protection from predators'

*accept 'more food' **or** 'they would have food' **or** 'partridges will come to feed' if none of the first three marking points are given*

*do **not** accept 'they had not been sprayed with chemicals'*

2 (L5)

[5]